

SHETH L.U.J AND SIR M.V COLLEGE

M2

7 Performing one-way ANOVA using aov() (R).

INPUT :-

```
library(dplyr)

# Load the dataset
df <- read.csv("world_top_restaurants_dataset.csv")

# View structure
str(df)

# Convert categorical variable to factor
df$Cuisine_Type <- as.factor(df$Cuisine_Type)

# One-Way ANOVA
anova_model <- aov(Average_Price_USD ~ Cuisine_Type, data = df)

# Display ANOVA table
summary(anova_model)

# Boxplot visualization
boxplot(
  Average_Price_USD ~ Cuisine_Type,
  data = df,
  col = c("lightblue", "lightgreen", "lightpink", "lightyellow"),
  main = "Average Price Comparison Across Cuisine Types",
  xlab = "Cuisine Type",
  ylab = "Average Price (USD)"
)
```

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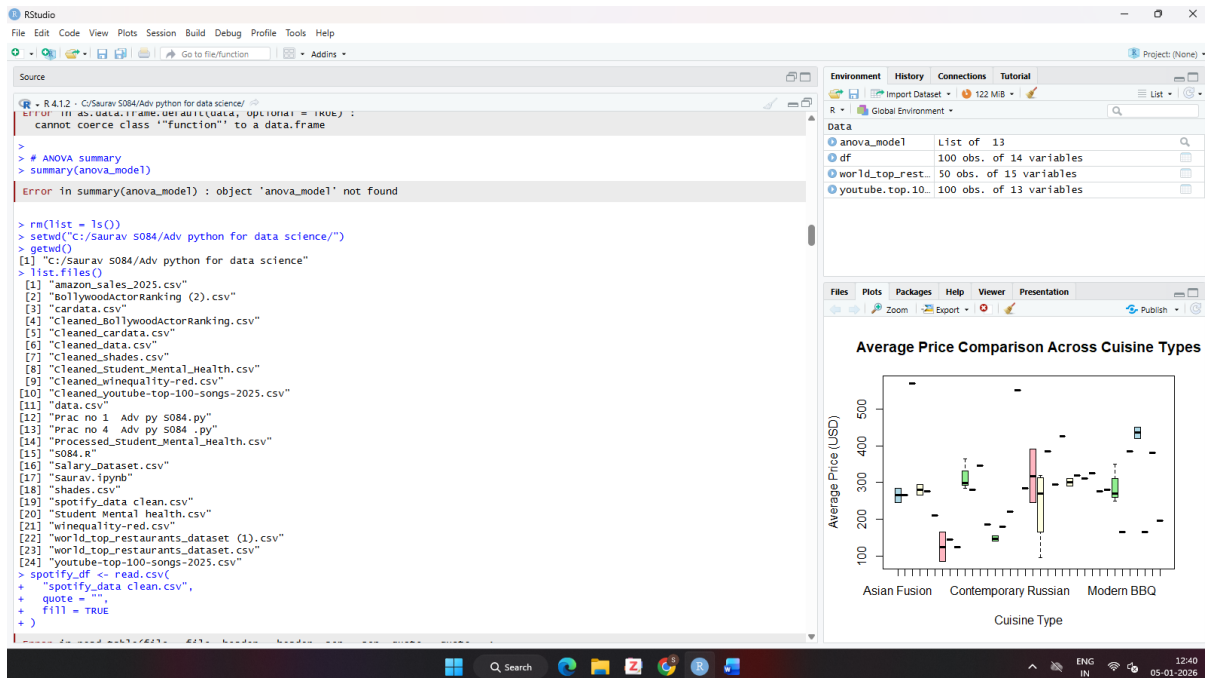
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Post-hoc test

TukeyHSD(anova_model)

OUTPUT :-



8 :- Performing two-way ANOVA using aov() (R).

INPUT :-

library(dplyr)

df <- read.csv("youtube-top-100-songs-2025.csv")

str(df)

df\$categories <- as.factor(df\$categories)

df\$Views_Group <- as.factor(

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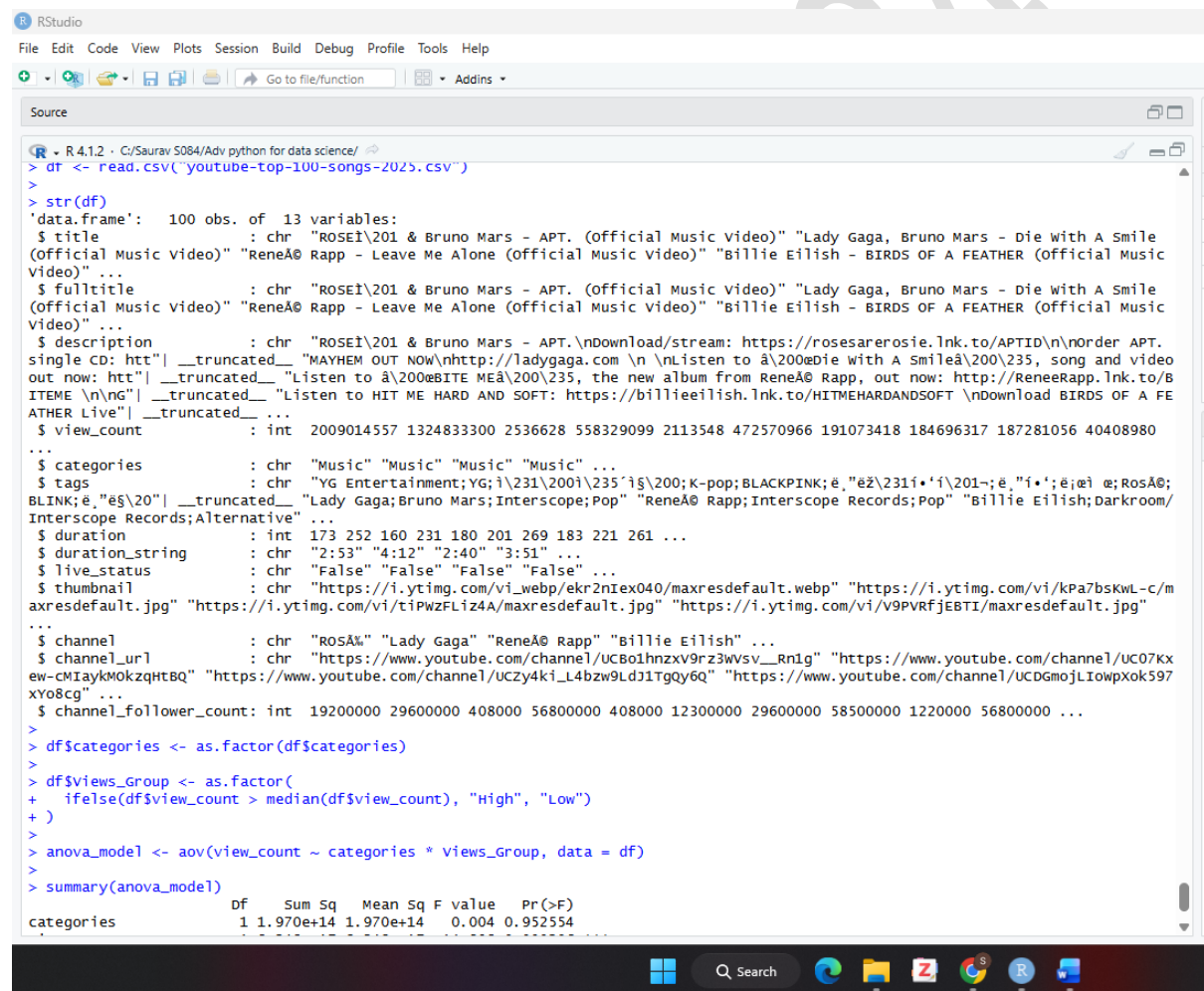
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```
ifelse(df$view_count > median(df$view_count), "High", "Low")
)
```

```
anova_model <- aov(view_count ~ categories * Views_Group, data = df)
```

```
summary(anova_model)
```

OUTPUT :-



```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins

Source
R • R 4.1.2 • C:/Saurav S084/Adv python for data science/
> df <- read.csv("youtube-top-100-songs-2025.csv")
>
> str(df)
'data.frame': 100 obs. of 13 variables:
 $ title      : chr "ROSEI\201 & Bruno Mars - APT. (Official Music Video)" "Lady Gaga, Bruno Mars - Die with A Smile (Official Music Video)" "ReneÅ Rapp - Leave Me Alone (Official Music Video)" "Billie Eilish - BIRDS OF A FEATHER (Official Music Video)" ...
 $ fulltitle   : chr "ROSEI\201 & Bruno Mars - APT. (Official Music Video)" "Lady Gaga, Bruno Mars - Die with A Smile (Official Music Video)" "ReneÅ Rapp - Leave Me Alone (Official Music Video)" "Billie Eilish - BIRDS OF A FEATHER (Official Music Video)" ...
 $ description : chr "ROSEI\201 & Bruno Mars - APT.\nDownload/stream: https://rosesarerosie.lnk.to/APTID\n\nOrder APT. single CD: htt| _truncated_ "MAYHEM OUT NOW\nhttp://ladygaga.com \n \nListen to à\200eDie with A Smileà\200\235, song and video out now: htt| _truncated_ "Listen to à\200eBITE MEà\200\235, the new album from ReneÅ Rapp, out now: http://ReneeRapp.lnk.to/BITEME \n\ng| _truncated_ "Listen to HIT ME HARD AND SOFT: https://billieeilish.lnk.to/HITMEHARDANDSOFT \nDownload BIRDS OF A FEATHER Live"| _truncated_ ...
 $ view_count  : int 2009014557 1324833300 2536628 558329099 2113548 472570966 191073418 184696317 187281056 40408980 ...
 $ categories  : chr "Music" "Music" "Music" "Music" ...
 $ tags        : chr "YG Entertainment;YG;\i\231\200\235\i§\200;K-pop;BLACKPINK;ë, "ëŽ\231f•'f\201-;ë, "f•';ë;æi e; RosÅ; BLINK;ë, "ë§\20"| _truncated_ "Lady Gaga;Bruno Mars;Interscope;Pop" "ReneÅ Rapp;Interscope Records;Pop" "Billie Eilish;Darkroom/Interscope Records;Alternative" ...
 $ duration    : int 173 252 160 231 180 201 269 183 221 261 ...
 $ duration_string : chr "2:53" "4:12" "2:40" "3:51" ...
 $ live_status  : chr "False" "False" "False" "False" ...
 $ thumbnail   : chr "https://i.ytimg.com/vi_webp/ekr2niex040/maxresdefault.webp" "https://i.ytimg.com/vi/kPa7bskWL-c/mxresdefault.jpg" "https://i.ytimg.com/vi/tiPwzFLiz4A/maxresdefault.jpg" "https://i.ytimg.com/vi/V9PVRfjEBTI/maxresdefault.jpg" ...
 $ channel     : chr "ROSÅ" "Lady Gaga" "ReneÅ Rapp" "Billie Eilish" ...
 $ channel_url  : chr "https://www.youtube.com/channel/UCB0ihnzxv9rz3wvsv_Rnlg" "https://www.youtube.com/channel/UC07KxeW-cMIaykMokzqHTBQ" "https://www.youtube.com/channel/UCzy4kiL4bzW9Ld31TgQy6Q" "https://www.youtube.com/channel/UCDgmojLIowpxok597xv08cg" ...
 $ channel_follower_count : int 19200000 29600000 408000 56800000 408000 12300000 29600000 58500000 1220000 56800000 ...
>
> df$categories <- as.factor(df$categories)
>
> df$Views_Group <- as.factor(
+   ifelse(df$view_count > median(df$view_count), "High", "Low")
+ )
>
> anova_model <- aov(view_count ~ categories * Views_Group, data = df)
>
> summary(anova_model)
      Df Sum Sq Mean Sq F value Pr(>F)
categories 1 1.970e+14 1.970e+14  0.004 0.952554
```

9 Conducting Chi-square tests using chisq.test() (R)

INPUT :-

```
library(dplyr)
```

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```
df <- read.csv("Salary_Dataset.csv")
```

```
str(df)
```

```
df$Salary_Group <- cut(  
  df$Salary,  
  breaks = c(min(df$Salary), 30000, 60000, max(df$Salary)),  
  labels = c("Low", "Medium", "High"),  
  include.lowest = TRUE  
)
```

```
table_gender_salary <- table(df$Gender, df$Salary_Group)
```

```
table_gender_salary
```

```
chi_result <- chisq.test(table_gender_salary)
```

```
chi_result
```

OUTPUT :-

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M2

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins

Source
R 4.1.2 C:/Saurav SO84/Adv python for data science/

---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> Salary_Dataset <- read.csv("C:/Saurav SO84/Adv python for data science/salary_Dataset.csv")
> view(Salary_Dataset)
> library(dplyr)
>
> df <- read.csv("Salary_Dataset.csv")
>
> str(df)
'data.frame':  1200 obs. of  17 variables:
 $ Employee_ID      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Name             : chr  "Employee_1" "Employee_2" "Employee_3" "Employee_4" ...
 $ Age              : int  58 48 34 27 40 58 38 42 30 30 ...
 $ Gender           : chr  "Female" "Female" "Male" "Male" ...
 $ Country          : chr  "India" "UK" "USA" "UK" ...
 $ City             : chr  "NY" "NY" "London" "Karachi" ...
 $ Education        : chr  "Bachelor" "High School" "High School" "Bachelor" ...
 $ Job_Title        : chr  "Manager" "Designer" "Manager" "Developer" ...
 $ Department       : chr  "IT" "HR" "IT" "Marketing" ...
 $ Experience_Years  : int  0 4 1 19 2 18 17 28 21 17 ...
 $ Salary_USD       : int  130735 54363 76694 31582 109910 96844 51301 104329 31620 92224 ...
 $ Bonus_USD        : int  3650 14600 15317 19768 18193 16563 7895 18543 2531 6481 ...
 $ Work_Hours_Per_Week: int  41 52 51 51 59 34 40 33 56 33 ...
 $ Remote_Work      : chr  "Yes" "No" "Yes" "Yes" ...
 $ Performance_Score : int  1 2 8 5 7 1 5 3 2 2 ...
 $ Joining_Year     : int  2015 2022 2010 2004 2008 2002 2007 2005 2003 2011 ...
 $ Contract_Type    : chr  "Part-Time" "Part-Time" "Part-Time" "Contract" ...

>
> df$Salary_Group <- cut(
+   df$Salary,
+   breaks = c(min(df$Salary), 30000, 60000, max(df$Salary)),
+   labels = c("Low", "Medium", "High"),
+   include.lowest = TRUE
+ )
>
> table_gender_salary <- table(df$Gender, df$Salary_Group)
>
> table_gender_salary

      Low Medium High
Female  0    166  430
Male    1    149  454

>
> chi_result <- chisq.test(table_gender_salary)
```

```
> df$Salary_Group <- cut(
+   df$Salary,
+   breaks = c(min(df$Salary), 30000, 60000, max(df$Salary)),
+   labels = c("Low", "Medium", "High"),
+   include.lowest = TRUE
+ )
>
> table_gender_salary <- table(df$Gender, df$Salary_Group)
>
> table_gender_salary

      Low Medium High
Female  0    166  430
Male    1    149  454

>
> chi_result <- chisq.test(table_gender_salary)

Warning message:
In chisq.test(table_gender_salary) :
  chi-squared approximation may be incorrect

>
> chi_result

Pearson's Chi-squared test

data:  table_gender_salary
X-squared = 2.5158, df = 2, p-value = 0.2842

> |
```

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